

Setup

The following code sets up the system by downloading the library from the web.

Make a beautiful flower

1. Make the first pedal look nice by reducing the domain.
2. Create the other pedals. Have fun coloring your pedals.
3. After you get the pedals, set `hide_axes=True` in `graph_funs()`.
4. Remove `plot.plot_funs()` function calls to plot one pedal at a time only.
5. Save each pedal into a different picture.

Here is a list of colors: "blue", "green", "red", "cyan", "magenta", "yellow", "black", "white", "olive", "indigo".

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In [3]: # MODIFY to add multiple transformations

# Define x as a symbolic variable
x = sp.symbols('x')

# Define exponential function
f1 = 0.25*(2)**(x)

# Create a graph object with equal_axes)
plot = graph_funs(equal_axes=True, hide_axes=True)

# Setup the grid
# Determine the range for X based on reflections and translations.
# Determine the range for Y based on reflections and translations (Look at maximum values).
plot.set_XY(minX=-10, maxX=+10, minY=-10, maxY=+10)

# Add to the list of functions:
domain1 = [0.33, 4, 100]

# Plot fun
plot.add_fun(f1, domain1, 'f1(x)='+str(f1), 'light blue') # Add the original function.
plot.plot_funs(plot_title="Original function f1 (light blue)", img_name="frame1.png")

# Add the inverse function.
domain3 = domain1
f3 = plot.add_inverse_fun(f1, domain3, 'inverse of f1', 'light blue')
plot.plot_funs(plot_title="f1, f2, inverse f1", img_name="frame2.png")

# Flip the inverse function:
f4 = -f3
domain4 = domain1
plot.add_fun(f4, domain4, 'f4', 'light blue')
plot.plot_funs(plot_title="f1, f2, f3, f4", img_name="frame3.png")

# Reflect f4:
f5 = f4.subs(x, -x)
domain5 = [-4, -0.33, 100]
plot.add_fun(f5, domain5, 'f5', 'light blue')
```

```

plot.plot_funs(plot_title="f1, f2, f3, f4, f5", img_name="frame4.png")

# Reflect f5:
f6 = -f5
domain6 = domain5
plot.add_fun(f6, domain6, 'f6', 'light blue')
plot.plot_funs(plot_title="f1, f2, f3, f4, f5, f6", img_name="frame5.png")

f8 = 0.25*(2)**(-x)
domain8 = [-4, -0.33, 100]
plot.add_fun(f8, domain8, 'f8', 'light blue')
plot.plot_funs(plot_title="f1-f8", img_name="frame6.png")

f9 = -0.25*(2)**(x)
domain9 = [0.33, 4, 100]
plot.add_fun(f9, domain9, 'f9', 'light blue')
plot.plot_funs(plot_title="f1-f9", img_name="frame7.png")

f10 = -0.25*(2)**(-x)
domain10 = [-4, -.33, 100]
plot.add_fun(f10, domain10, 'f10', 'light blue')
plot.plot_funs(plot_title="f1-f10", img_name="frame8.png")

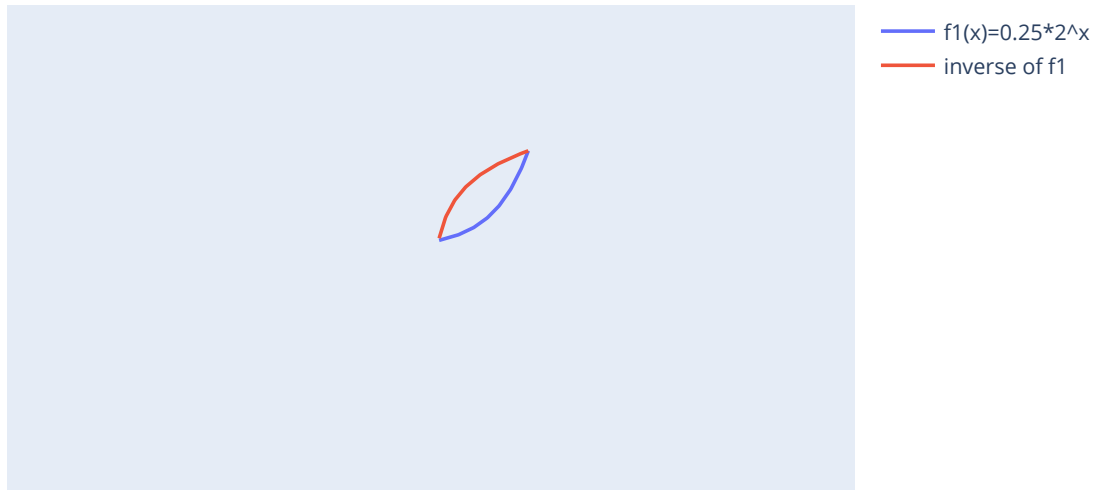
#plot.plot_funs(plot_title="f1, f2, inverse f1", img_name="frame1.png")
#plot.plot_funs(plot_title="f1-f8", img_name="frame2.png")
#plot.plot_funs(plot_title="f1-f9", img_name="frame4.png")

```

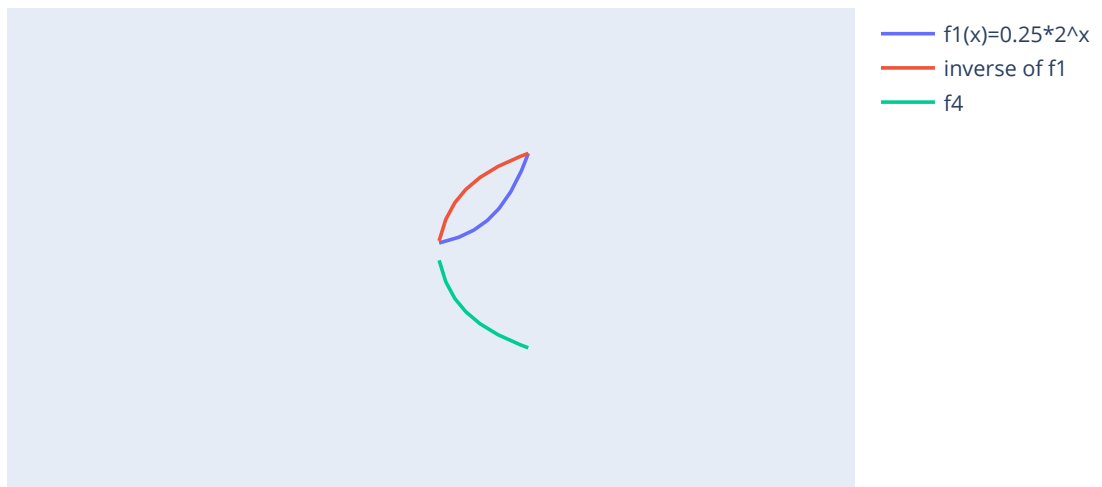
Original function f1 (light blue)



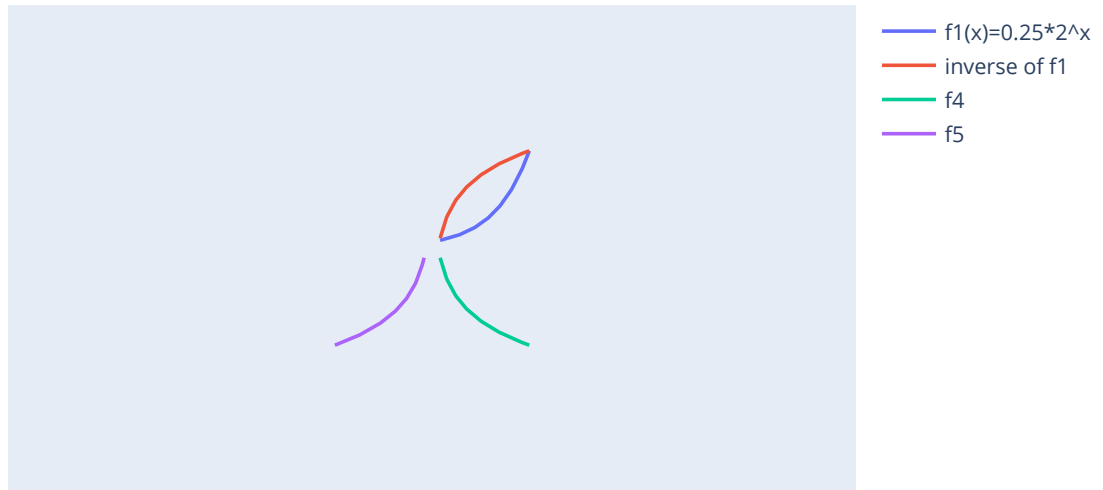
f1, f2, inverse f1



f1, f2, f3, f4



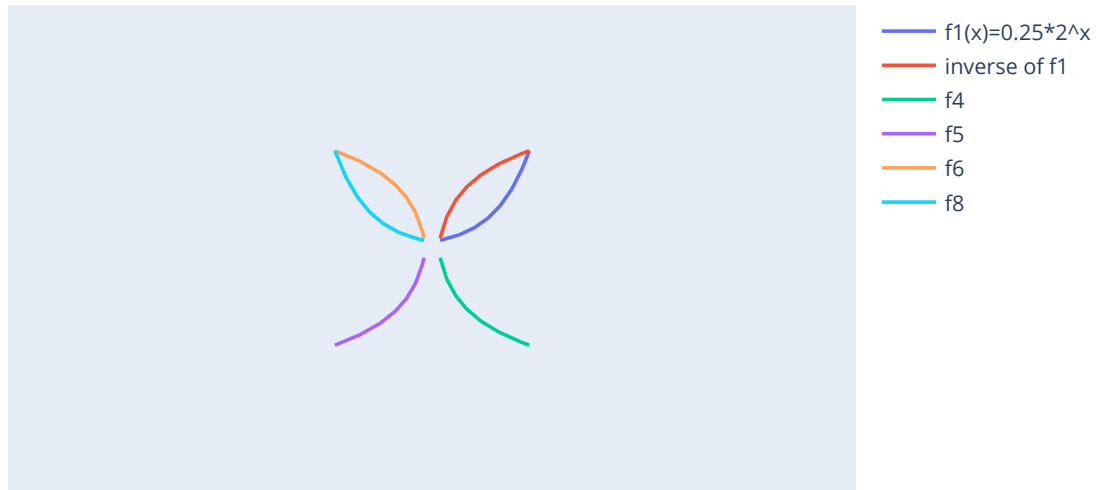
f1, f2, f3, f4, f5



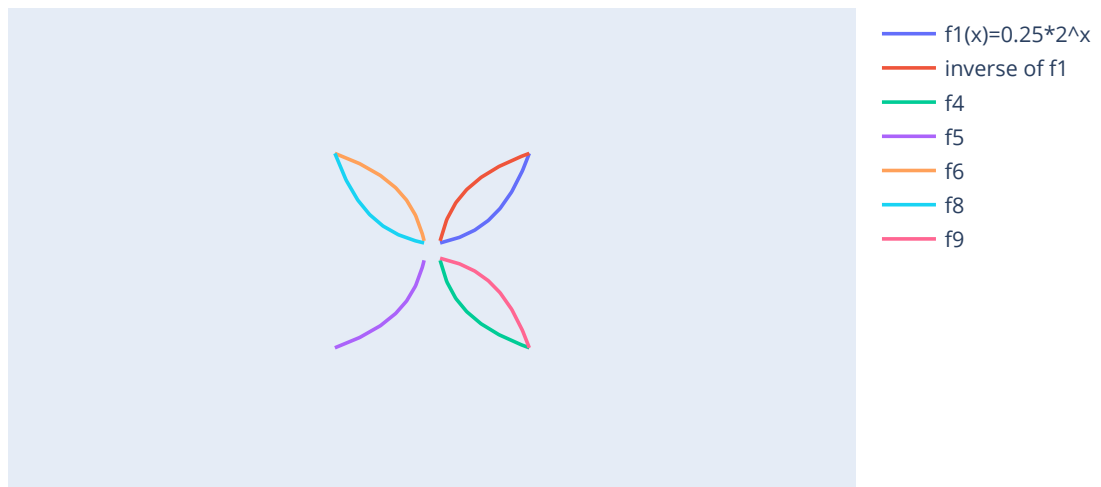
f1, f2, f3, f4, f5, f6



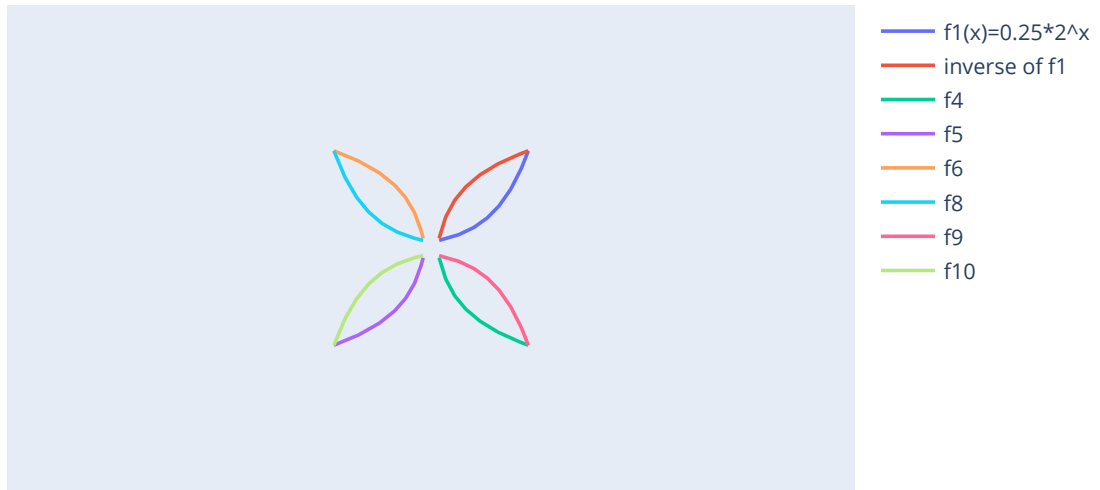
f1-f8



f1-f9



f1-f10



```
In [11]: !jupyter nbconvert --to html /content/drive/MyDrive/Colab\ Notebooks/Used_for_PDF_Graphfuns_Fl  
[NbConvertApp] Converting notebook /content/drive/MyDrive/Colab Notebooks/Used_for_PDF_Graphfun  
s_Flower_Project_C3_G3.ipynb to html  
[NbConvertApp] Writing 518358 bytes to /content/drive/MyDrive/Colab Notebooks/Used_for_PDF_Grap  
hfuns_Flower_Project_C3_G3.html
```